MODULE 3 CARDIAC CONDUCTION

The Cardiac Conduction System.

BLOCK 3 · MODULE 3: CONDUCTION AND DISORDERS

A reference for the cardiac conduction video and lab. This chapter covers the conduction pathway and its components, the nerve supply to the heart, what an electrocardiogram records, and common conduction disorders.

BY THE END

1. Trace the conduction pathway from the SA node to the Purkinje fibers.
2. Locate each component of the conduction system and state its role.
3. Describe the nerve supply to the heart and what an ECG records.
4. Name common conduction disorders.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Cardiac conduction

drsrennie-stack.github.io/new-build-bio4-solano/conduction-concept-videos.html

The Conduction System, an Overview

The heart sets its own beat. A network of specialised cardiac muscle cells, the conduction system, generates each heartbeat and spreads it so the chambers contract in the right order.

WHAT THE SYSTEM IS

Structure	What it is
Cardiac conduction system	The network of specialised cardiac muscle cells that generates and spreads the signal for each heartbeat. About one percent of the cardiac fibers
Autorhythmicity	The heart's built-in ability to generate its own rhythm without a signal from the nervous system
Conducting cells	Specialised non-contractile cardiac cells that generate and carry the electrical signal
Contractile cells	The ordinary working cardiomyocytes that contract when the signal reaches them
Why the order matters	The conduction system makes the atria contract first and the ventricles second, so each chamber fills before it empties

The Conduction Pathway

Each heartbeat travels a fixed route through the conduction system. Follow the signal in order, from the pacemaker to the ventricle walls.

1. **SA node** the sinoatrial node, the pacemaker; it fires first and sets the pace of the whole heart
2. **Across the atria** the signal spreads through both atria, and the atria contract
3. **AV node** the atrioventricular node; the signal pauses here briefly, giving the atria time to finish emptying
4. **AV bundle** the atrioventricular bundle, or bundle of His; it carries the signal into the interventricular septum, the only electrical link between the atria and the ventricles

5. **Bundle branches** the right and left bundle branches carry the signal down both sides of the interventricular septum toward the apex
6. **Purkinje fibers** spread the signal through the walls of both ventricles, and the ventricles contract from the apex upward

The Components of the Conduction System

COMPARE EACH BY WHERE IT SITS AND WHAT IT DOES

Structure	Location	Role
SA node	The wall of the right atrium, near the opening of the superior vena cava	The natural pacemaker; it fires fastest, about sixty to one hundred times a minute, and sets the heart rate
AV node	The floor of the right atrium, near the interatrial septum	Delays the signal briefly, then passes it on to the ventricles
AV bundle	The superior part of the interventricular septum	The only electrical bridge from the atria to the ventricles
Bundle branches	The right and left sides of the interventricular septum	Carry the signal down toward the apex of the heart
Purkinje fibers	Throughout the walls of both ventricles	Deliver the signal to the contractile cells of the ventricle walls

HOLD ONTO THIS

The fibrous skeleton insulates the atria from the ventricles, so the AV bundle is the only route through. That single bridge is why a block at the AV node or bundle disconnects the two halves of the heartbeat entirely.

Nerve Supply to the Heart

The conduction system runs on its own, but the nervous system can speed it up or slow it down. Autonomic nerves reach the SA and AV nodes and adjust the rate to the body's needs.

THE NERVES THAT MODULATE THE RATE

Structure	What it is
Cardiac centers	Regions of the medulla oblongata in the brainstem that adjust the heart rate
Cardiac plexus	The network of autonomic nerve fibers near the base of the heart through which these nerves reach the nodes
Sympathetic nerves	Autonomic nerves that reach the heart through the cardiac plexus and speed the heart rate
Parasympathetic nerves	Autonomic fibers carried in the vagus nerve that slow the heart rate
A modulating role	These nerves do not start the heartbeat; they only adjust the pace the SA node already sets

HOLD ONTO THIS

A transplanted heart has had its nerves cut and it still beats. That is the cleanest proof that autorhythmicity belongs to the conduction system and not to the nervous system.

The Electrocardiogram

An electrocardiogram, an ECG or EKG, is a tracing of the heart's electrical activity recorded from the body surface. Each deflection of one heartbeat lines up with an event in the conduction system.

COMPARE THE THREE DEFLECTIONS

Deflection	Conduction event	What follows
P wave	The signal spreads across the atria	The atria contract
QRS complex	The signal spreads across the ventricles	The ventricles contract
T wave	The ventricles electrically recover	The ventricles relax

Conduction Disorders

When the conduction system fails, the rhythm goes wrong. Compare the common disorders by what happens and why it matters.

THE COMMON CONDUCTION DISORDERS

Disorder	What happens	Why it matters
Arrhythmia	Any deviation from the normal heart rhythm	The umbrella term for the disorders below
Tachycardia	An abnormally fast resting heart rate	The chambers may not have time to fill between beats
Bradycardia	An abnormally slow resting heart rate	The body may not receive enough blood flow
Heart block	The signal is delayed or stopped at the AV node	The atria and ventricles can beat out of step
Atrial fibrillation	Rapid, disorganised signals in the atria	The atria quiver instead of contracting; common and treatable
Ventricular fibrillation	Rapid, disorganised signals in the ventricles	The ventricles cannot pump; a life-threatening emergency
Ectopic pacemaker	A site outside the SA node takes over the rhythm	The normal pacing order of the heart is lost

WHEN THE RHYTHM NEEDS HELP

An **artificial pacemaker** is a small implanted device that delivers a timed electrical signal to the heart. It takes over the job of a failing **SA node** or carries the signal past a block at the **AV node**, restoring an ordered beat.

In **ventricular fibrillation** the ventricles only quiver, so they move no blood. A defibrillator delivers a shock that stops all the chaotic signals at once, giving the SA node a chance to restart the heart in its normal rhythm.

The brief delay at the **AV node** is not a flaw, it is essential timing. It holds the signal just long enough for the atria to finish emptying into the ventricles before the ventricles contract, and an **ECG** is how a clinician reads whether that timing is intact.

The **SA node** is supplied by the right coronary artery in most people. That is why an occlusion of the RCA can produce a rhythm problem as well as a wall of dying muscle.